PROJECT REPORT

Medical Image Enhancement and Diagnostic Assistance System for Brain Tumor Detection

Brain MRI Enhancement, Tumor Localisation and Classification using OpenCV, Classical Machine Learning and a Fine-tuned CNN

Course: DIGITAL SIGNAL PROCESSING (DSP) · **Group 11**

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1. Abstract

This project develops a complete software system that assists in detecting brain tumors on magnetic resonance imaging (MRI), from a raw slice to a verdict a clinician can check. Seven stages are integrated into one pipeline: acquisition, denoising by median and bilateral filtering, contrast enhancement by CLAHE with unsharp masking, segmentation of a suspicious region by skull stripping, a tissue-relative statistical threshold and a left–right asymmetry test, extraction of 171 engineered features, and classification. Detection (tumor or no tumor) uses a support vector machine on the engineered features alone, so its evidence remains inspectable; tumor typing (glioma, meningioma, pituitary) adds a fine-tuned ResNet-18 because the classical features measurably fall short there. Every stage is shown to the user through a REST API and a web interface. On the public dataset's held-out test split the system reaches 95.4% detection accuracy with 93.8% sensitivity and 80.2% typing accuracy. A duplicate audit we ran on that split then shows that 259 of its 394 test slices are near-copies of training slices; on the 135 slices without a copy, detection sensitivity is 86.7% and typing accuracy 43.7%. Both sets of numbers are reported, because the second is the honest estimate of the first.

2. Introduction and Motivation

MRI is the standard imaging modality for brain tumors, but the images carry noise, low contrast and unclear boundaries, and reading them is slow and tiring even for trained radiologists. Magnitude MRI noise is Rician rather than Gaussian, which biases dark regions upwards and makes low-contrast lesions harder to separate from tissue[4]. Published classifiers on public brain-tumor datasets now routinely report accuracy above 95%, yet a bare label with a confidence score gives a radiologist nothing to verify.

This project therefore treats the problem as a signal-processing chain rather than a classifier in isolation. The image is first cleaned and enhanced, then a suspicious region is located and measured, and only then classified; each intermediate result is exposed so that a clinician can see where the system looked and overrule it. The second aim is measurement: every claim (enhancement gain, segmentation behaviour, accuracy) is measured on data the model did not train on, and the measurement method itself is checked.

3. Stage-by-Stage Comparison with the Source Literature

The system is built as seven stages, each grounded in an established method from the literature. Stages 2–5 are deterministic signal and image processing; Stage 6 learns from data; Stage 7 presents the result. Table 1 states, for each stage, the published work it is taken from, what was actually built here, and the outcome this prototype measured for it.

Table 1. The pipeline stages against their basis in the literature, with the measured outcome for each.

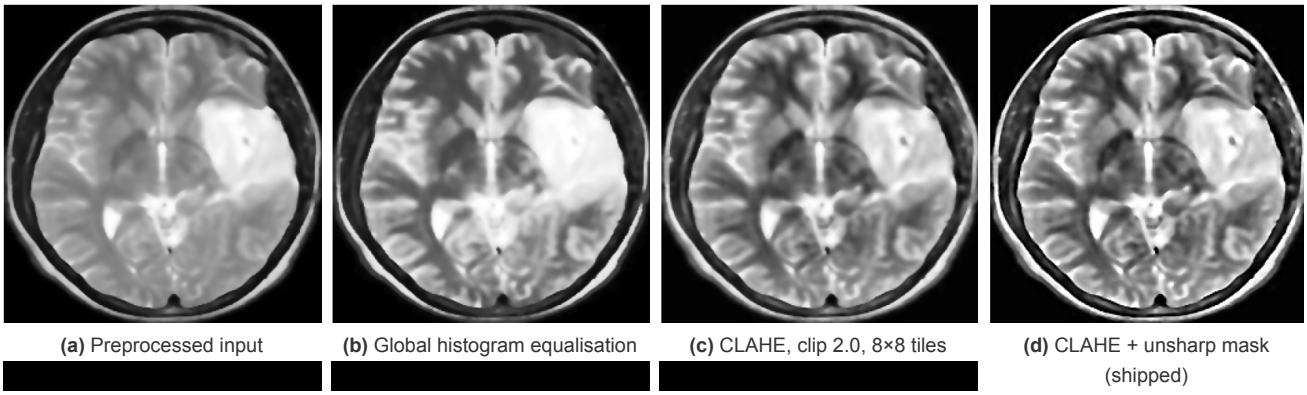
Stage	Basis in the literature	What this project built	Measured outcome
S1 Data	Bhuvaji et al. 4-class brain MRI corpus[10]; Chakrabarty binary set[11].	Loaders for both, using the curators' own train/test split for the 4-class set (2,870 / 394 slices), plus an MRI phantom generator (skull, CSF, grey/white matter, ventricles, tumor with edema, B1 bias field, Rician noise) for reproducible tests.	3,264 slices: 500 no tumor, 926 glioma, 937 meningioma, 901 pituitary. The test split is audited in Section 7.

S2 Preprocessing	Rician noise model of magnitude MRI[4]; bilateral filtering[3]; median filtering for impulse noise.	Grayscale, blank-border crop, resize to 256×256 (area interpolation when shrinking), median 3×3, bilateral (d = 7, $\sigma_{colour} = \sigma_{space} = 50$), min–max normalisation.	Stages 2–5 together take 0.026 s per slice on a laptop CPU.
S3 Enhancement	Adaptive histogram equalisation[2] and its contrast-limited form, CLAHE[1].	CLAHE (clip limit 2.0, 8×8 tiles) then unsharp masking (σ = 1.5, amount 1.2); global equalisation kept for comparison; contrast, entropy and sharpness measured per image.	Over the 394 test slices: entropy 6.77 → 7.39 bits, sharpness 338 → 1428, contrast 57.9 → 70.4 (Table 2).
S4 Segmentation	Otsu thresholding[5] for the head mask; distance-transform skull peel; mathematical morphology.	Skull strip, then flag tissue ≥ mean + 1.5σ (over pixels above the brain median), union with a mirrored-asymmetry mask (≥ mean + 2σ), 5×5 opening ×2 and closing ×3, contours ≥ 120 px.	Otsu inside the brain flagged 100% of healthy brains on 60 scans; the statistical rule 3%. 129 of 155 tumors localised on the 253-image set.
S5 Features	Grey-level co-occurrence texture[6]; local binary patterns[7].	171 features in seven families: intensity + histogram 33, GLCM 38, LBP 28, shape 12, asymmetry 10, regional 20, location + plane 30.	Texture-only typing reached 72.1%; the location family was added because tumor type follows position.
S6 Classification	Support-vector networks[8]; deep residual networks[9].	Four classical models compared by 5-fold cross-validated ROC-AUC; sensitivity-first threshold tuning on out-of-fold predictions; ResNet-18 fine-tuned for typing and blended with a weight chosen on a validation split.	Detection: 95.4% accuracy, 93.8% sensitivity, 100% specificity, AUC 0.997. Typing: 80.2% (classical alone 69.3%).
S7 Presentation	Radiology decision support: the system must show its evidence, not only a label.	An 8-panel figure, a FastAPI service returning every stage image and measurement, and a Next.js web app. Uploaded scans are never written to disk.	66 automated tests across the pipeline, location features, CNN branch and API.

> **The design constraint that ties the table together.** Detection is decided by classical features only, so the region, its area, circularity, solidity, brightness and location are all visible and checkable. The CNN is added only to typing, where the classical model was measured to fall short (69.3% against 80.2%), and the interface states which branch produced each answer.

4. Enhancement — Results

Stage 3 is evaluated with three image-quality measures computed by the project's own `enhancement_report()`: contrast as the standard deviation of intensities, Shannon entropy of the 256-bin histogram in bits, and sharpness as the variance of the Laplacian. Table 2 reports their mean over the 394 test slices for each enhancement method; Figure 1 shows the methods on one slice.



(a) Preprocessed input | **(b)** Global histogram equalisation | **(c)** CLAHE, clip 2.0, 8×8 tiles | **(d)** CLAHE + unsharp mask (shipped)

Figure 1. One glioma slice through each enhancement method. Global equalisation applies a single mapping driven by the whole histogram, in which background and bright skull dominate; CLAHE equalises each 32×32-pixel tile with a clip limit that stops noise in flat regions from being amplified; unsharp masking then restores edge detail.

Table 2. Mean image-quality measures over the 394 test slices. Best in each row in blue.

Measure	Preprocessed	Global HE	CLAHE only	CLAHE + unsharp
Contrast (std of intensities)	57.9	**75.6**	65.3	70.4
Entropy (bits)	6.77	6.55	**7.41**	7.39
Sharpness (Laplacian variance)	338	808	391	**1428**

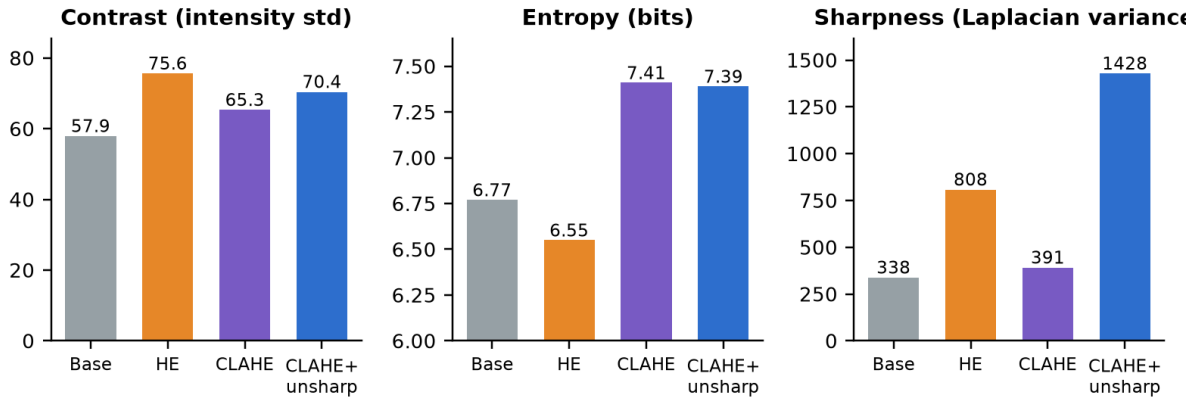


Figure 2. The same measures as bars. Global equalisation produces the largest raw contrast but lowers entropy below that of its input, i.e. it merges grey levels and discards information. CLAHE raises entropy, and unsharp masking adds a four-fold gain in sharpness. The shipped chain improved contrast on 98.5%, entropy on 99.0% and sharpness on 100% of the test slices.

These are image-quality measures, not diagnostic ones. Kondrateva et al.[12] show for deep segmentation networks that intensity steps which make images look better (bias-field correction, denoising, histogram matching) give no significant downstream gain and can reduce it. A with/without-CLAHE ablation on the classifiers is therefore listed as future work rather than assumed.

5. Detection — Results

Every slice is reduced to the same 171 features, and four classical models are compared by 5-fold stratified cross-validation on the training split only, with the feature scaler inside each fold so that no scaling information leaks across folds. The SVM with an RBF kernel ($C = 10$) is selected by cross-validated ROC-AUC (Figure 3).

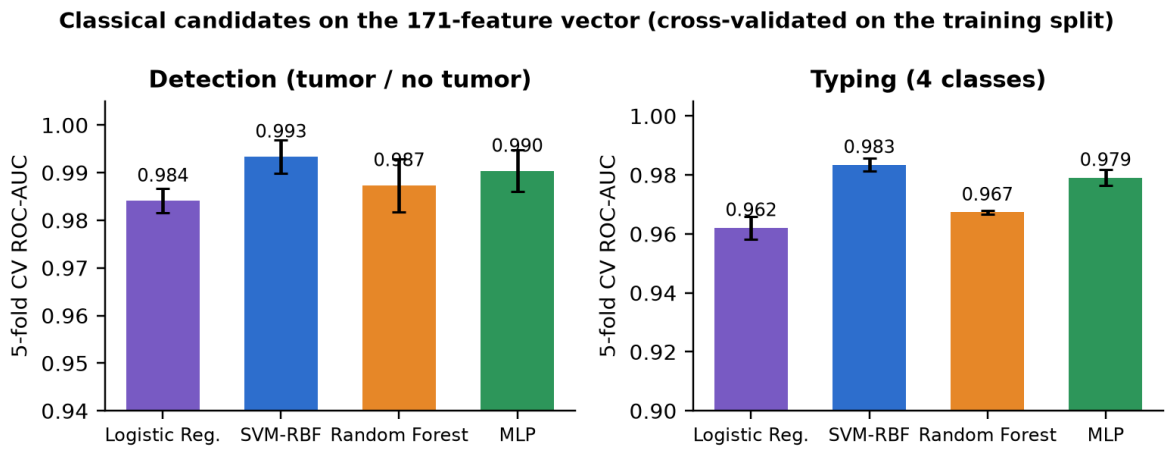


Figure 3. Cross-validated ROC-AUC (mean ± std over 5 folds) of the four candidates on the training split, for detection and for typing. SVM-RBF is selected for both tasks (0.993 and 0.983).

Because a missed tumor costs far more than a second look, the decision threshold is tuned for sensitivity: it is the lowest threshold that keeps the out-of-fold specificity on the training split at or above 80%, which gives 0.153. The test set is never used to choose it. Table 3 compares the default operating point with the tuned one.

Table 3. Detection on the 394 held-out test slices (289 tumor, 105 healthy), SVM-RBF on 171 features.

Operating point	Accuracy	Balanced acc.	Sensitivity	Specificity	Missed tumors	False alarms	ROC-AUC
Default	91.6%	94.3%	88.6%	100%	33	0	0.997
Tuned, threshold 0.153	**95.4%**	**96.9%**	**93.8%**	100%	**18**	0	0.997

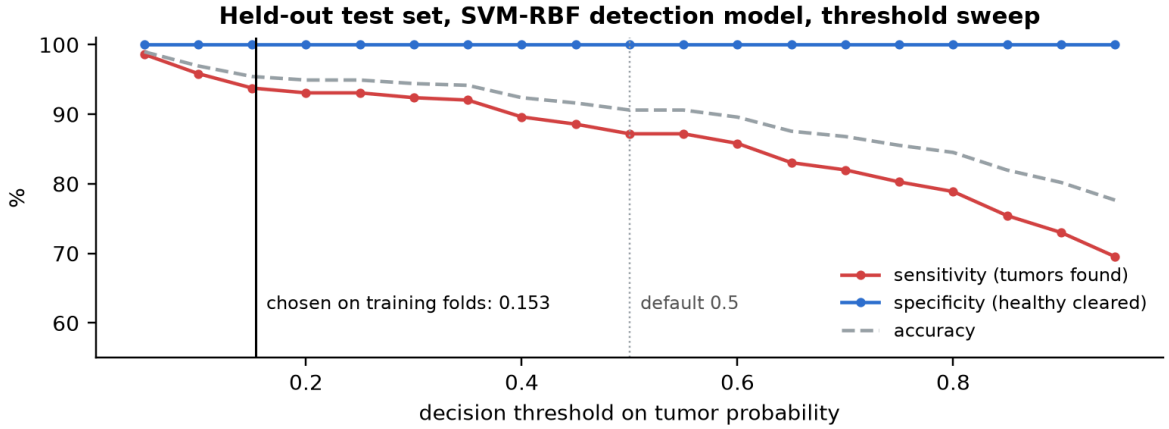


Figure 4. Sensitivity, specificity and accuracy on the test set as the threshold moves (recorded for transparency, not used for selection). Specificity stays at 100% at every threshold; Section 7 explains why. Note: the default row in Table 3 uses the SVM's own decision function, which matches the 0.45 row of this sweep rather than 0.50.

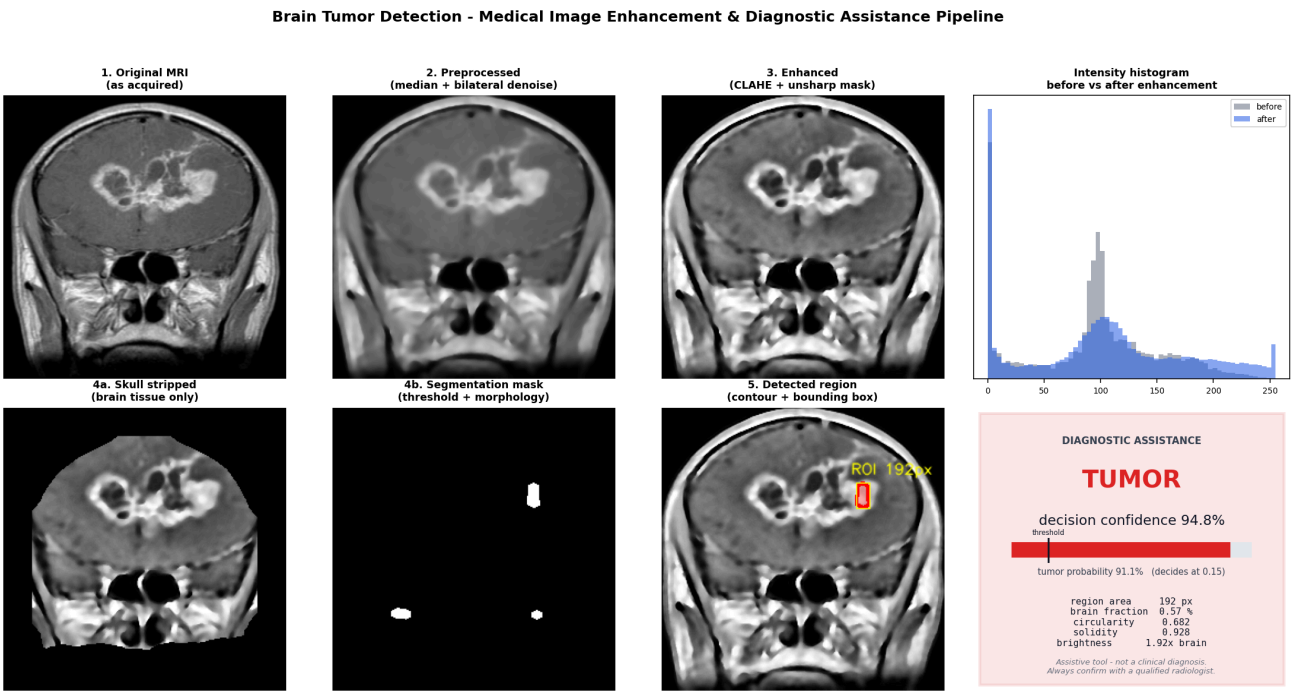


Figure 5. What the user sees for one coronal slice: original, denoised, enhanced and its histogram, skull-stripped brain, segmentation mask, detected region, and the verdict panel with the tumor probability against the threshold and the region's measurements. The outlined region (192 px) covers only one bright part of the enhancing lesion; the verdict comes from all 171 features rather than from the box, and showing the box is what lets a clinician see when it is wrong.

6. Typing — Results

The same classical pipeline was trained to name the tumor type, and reached 69.3% test accuracy; glioma and pituitary recall were 23% and 46%. Tumor type depends on spatial structure that a global texture histogram averages away, so a ResNet-18 pretrained on ImageNet is fine-tuned on the enhanced slices (224×224, all layers, 14 epochs, AdamW, label smoothing 0.05, augmentation, four-view test-time averaging), and its probabilities are blended with the SVM's. The blend weight is chosen on a 431-slice validation split carved out of the training folder, with 0 and 1 in the search grid so that the search also chooses the branch; it selected 0.95 on the CNN (Figure 8).

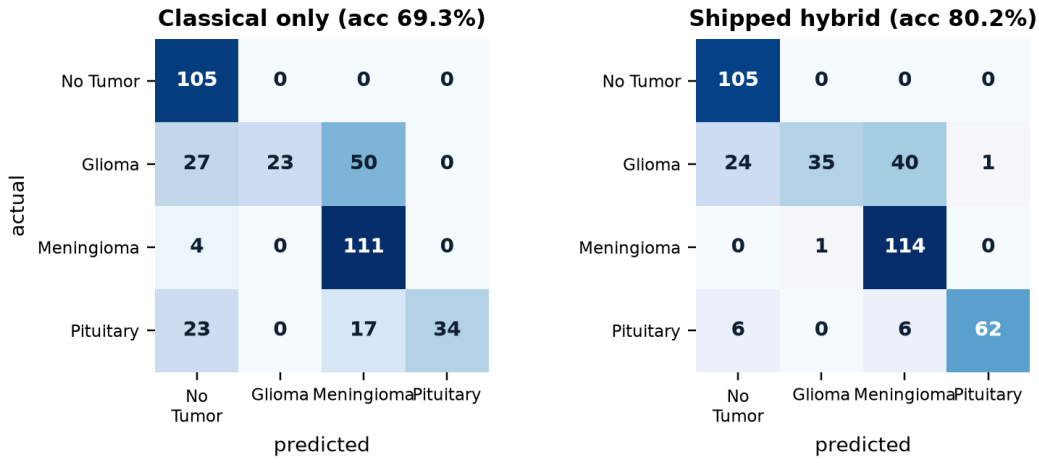


Figure 6. Test-set confusion matrices for the classical model alone and for the shipped blend. Of 100 test gliomas the blend calls 40 meningioma and 24 "no tumor".

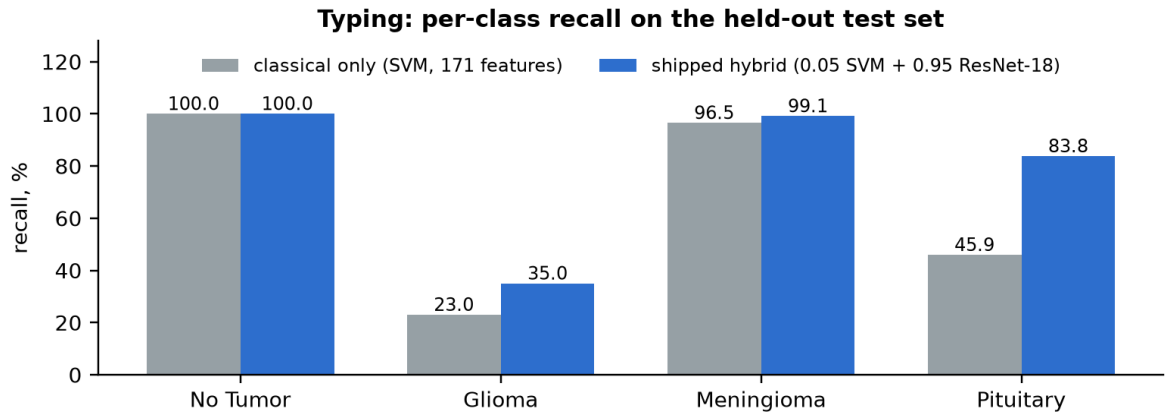


Figure 7. Per-class recall. The CNN branch lifts pituitary from 45.9% to 83.8% and glioma from 23.0% to 35.0%; no-tumor and meningioma are near-perfect for both. Macro F1 of the blend is 78.1% and its one-vs-rest AUC 0.961.

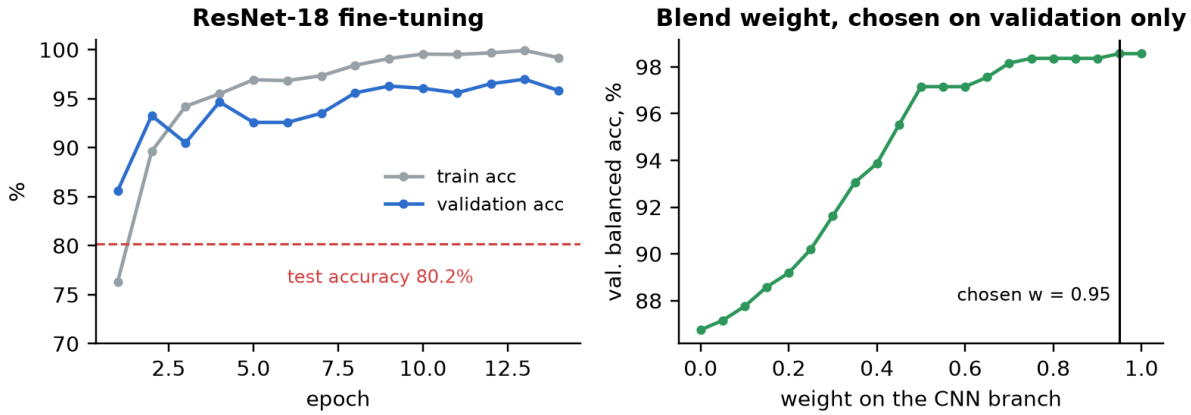


Figure 8. Left: CNN fine-tuning; the epoch with the best validation balanced accuracy (epoch 13, 97.4%) is restored at the end. Right: validation balanced accuracy against the blend weight. Validation and test differ sharply: 97.4% on validation slices against 79.5% balanced accuracy on the test folder. Every training glioma is 512×512 pixels while the 100 test gliomas come in 64 different sizes, so the two folders do not come from the same distribution.

7. Auditing the Test Split

The 100% specificity at every threshold and the near-perfect meningioma recall prompted a check of the test split itself. Following the criterion of Vangala et al.[13], every image was reduced to a 64×64 grey array, mean-centred and normalised to unit length, so that the inner product of two images is their zero-mean normalised cross-correlation ϱ. Each test slice was compared with all 2,870 training slices, and a slice with $\varrho \geq 0.98$ to any of them was counted as having a training twin. MD5 hashes were also compared to find byte-identical files.

The result: **88 test files are byte-identical to training files** (all of them healthy), and **259 of the 394 test slices have a twin at $\varrho \geq 0.98$**. The count is the same at every threshold from 0.98 to 0.995, so these are true copies, and it matches the 259 that Vangala et al. report for this dataset. The shipped models were then re-run on all 394 slices (reproducing Table 3 and Figures 6–7 exactly) and the results split by twin status.

Table 4. The shipped models on test slices with and without a training twin (ρ ≥ 0.98).

Measure	Twin in training (259)	No twin (135)	All (394)
Composition	105 healthy, 109 men., 25 pit., 20 gli.	0 healthy, 6 men., 49 pit., 80 gli.	105 / 115 / 74 / 100
Detection sensitivity	100% (154/154)	**86.7% (117/135)**	93.8%
Detection specificity	100% (105/105)	not measurable	100%
Typing accuracy	99.2%	**43.7%**	80.2%
Glioma recall	90.0% (18/20)	**21.3% (17/80)**	35.0%
Pituitary recall	100% (25/25)	75.5% (37/49)	83.8%
Meningioma recall	100% (109/109)	83.3% (5/6)	99.1%

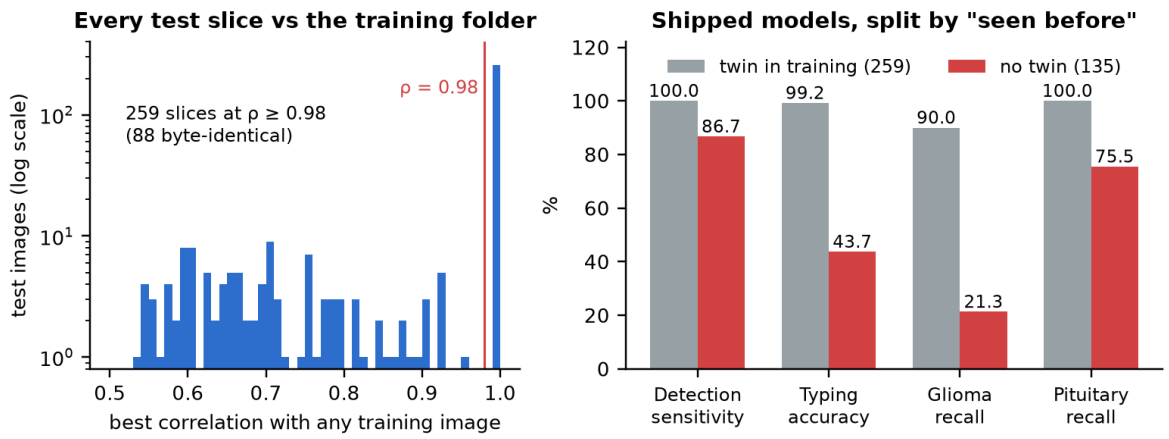


Figure 9. Left: each test slice's best correlation with any training slice (log scale); 259 sit at ρ ≈ 1. Right: the shipped models on slices with and without a training twin.

> **The headline result.** All 18 missed tumors are slices with no twin in training, and every healthy test slice has one. On the 135 slices the models have never seen anything like, detection sensitivity is 86.7% and typing accuracy 43.7%. Glioma is weak above all because 80 of the 100 test gliomas are genuinely new. The 100% specificity cannot be claimed for healthy brains outside the training data: none were in the test set. The standard numbers in Sections 5 and 6 are reported because they are what the literature reports on this dataset; Table 4 is the better estimate of how the system would behave on new scans.

Two of the project's own documents state that slices from one patient never appear on both sides of the split; this audit shows that assumption does not hold for this dataset and the statements should be corrected.

8. Engineering Findings

Four defects found during development changed the design, and each is now covered by a regression test. **(i)** When the head touches the top and bottom image borders, the background splits into two regions; a corner flood fill reached only one, and the other was filled in as "brain". Padding a one-pixel black frame before flood filling fixed it. **(ii)** Otsu's method always splits a histogram into two classes, so inside a healthy brain it returned grey versus white matter as a lesion, flagging 100% of healthy scans; the tissue-relative mean + 1.5σ rule flags 3%. **(iii)** The first threshold tuner maximised specificity under

a sensitivity floor and picked 0.93 from over-confident training folds, dropping test sensitivity from 91% to 71%; inverting the objective fixed it. **(iv)** The first ensemble kept whichever branch scored best on the test set, which reports the best of three draws; the blend weight is now chosen on a validation split.

9. Conclusions and Limits

The prototype delivers the system the synopsis proposed. A working pipeline runs end to end, from an uploaded slice to an annotated verdict, through a REST API and a web interface. Enhancement is measured: CLAHE with unsharp masking raises entropy and sharpness on almost every test slice, where global equalisation loses information. Detection on engineered features reaches 95.4% accuracy and 93.8% sensitivity on the dataset's test split, and typing reaches 80.2% with a CNN branch that the classical features alone could not match.

The limits are equally part of the result. The public test split shares 259 near-identical slices with the training split, and on the remaining 135 slices detection sensitivity is 86.7% and typing accuracy 43.7%; glioma typing is unsolved. There are no patient identifiers, so splits are by slice. The CNN branch that carries typing is not interpretable. The system works on 2-D slices, its intensity rules can be misled by calcification or artefacts, and it is an assistive tool, not a medical device. The next steps are a de-duplicated, patient-level evaluation, preferably on a second dataset such as that of Cheng et al.[14]; Grad-CAM over the CNN branch; a U-Net segmentation with Dice scoring; and a with/without ablation of each enhancement step.

10. References

1. K. Zuiderveld, "Contrast limited adaptive histogram equalization," in *Graphics Gems IV*, P. S. Heckbert, Ed. San Diego, CA, USA: Academic Press, 1994, pp. 474–485.
2. S. M. Pizer, E. P. Amburn, J. D. Austin, R. Cromartie, A. Geselowitz, T. Greer, B. ter Haar Romeny, J. B. Zimmerman, and K. Zuiderveld, "Adaptive histogram equalization and its variations," *Comput. Vis. Graph. Image Process.*, vol. 39, no. 3, pp. 355–368, 1987.
3. C. Tomasi and R. Manduchi, "Bilateral filtering for gray and color images," in *Proc. 6th Int. Conf. Computer Vision (ICCV)*, 1998, pp. 839–846.
4. H. Gudbjartsson and S. Patz, "The Rician distribution of noisy MRI data," *Magn. Reson. Med.*, vol. 34, no. 6, pp. 910–914, 1995.
5. N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Syst., Man, Cybern.*, vol. 9, no. 1, pp. 62–66, 1979.
6. R. M. Haralick, K. Shanmugam, and I. Dinstein, "Textural features for image classification," *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-3, no. 6, pp. 610–621, 1973.
7. T. Ojala, M. Pietikäinen, and T. Mäenpää, "Multiresolution gray-scale and rotation invariant texture classification with local binary patterns," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 7, pp. 971–987, 2002.
8. C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.
9. K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.
10. S. Bhuvaji, A. Kadam, P. Bhumkar, S. Dedge, and S. Kanchan, "Brain tumor classification (MRI)," Kaggle, 2020, doi: 10.34740/KAGGLE/DSV/1183165.
11. N. Chakrabarty, "Brain MRI images for brain tumor detection," Kaggle, 2019.
12. E. Kondrateva, P. Druzhinina, A. Dalechina, S. Zolotova, A. Golanov, B. Shirokikh, M. Belyaev, and A. Kurmukov, "Negligible effect of brain MRI data preprocessing for tumor segmentation," *Biomed. Signal Process. Control*, vol. 96, Art. no. 106599, 2024, doi: 10.1016/j.bspc.2024.106599.
13. B. P. Vangala, S. Guda, L. Peddi, and N. Vangala, "Scores that hold, benchmarks that leak: Measuring dataset contamination in public brain-tumor MRI classification," arXiv preprint arXiv:2610.00421, 2026.
14. J. Cheng, W. Huang, S. Cao, R. Yang, W. Yang, Z. Yun, Z. Wang, and Q. Feng, "Enhanced performance of brain tumor classification via tumor region augmentation and partition," *PLOS ONE*, vol. 10, no. 10, p. e0140381, 2015.